

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2022

Bachelor in Computer Applications

Course Title: **Network Programming**

Code No: **CACS355** Semester: **VI**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. What is the Client-Server software model? Explain the strengths and weaknesses of Java programming as a network programming language.
3. What is the use of InetAddress class? Write a program to retrieve IP and MAC addresses?
4. Define URL class. Write a program to show the parts of the URL (Splitting URL).
5. Define Socket. Differentiate between TCP and UDP Sockets with their applications.
6. Define cookies. Write a program to retrieve cookie information stored in the system.
7. Define IP Multicast. Write a program to join a computer system in a multicast group.
8. Write Short Notes on (Any Two): a) HTTP b) RMI c) Secure Socket

Group C

Attempt any TWO questions [2x10=20]

9. Write multithreaded TCP client and server socket programs for daytime service.
10. Write UDP client and server socket programs in which the server identifies the number sent by a client is either even or odd and replies to the client accordingly.
11. How blocking I/O is different from Non-Blocking I/O? Explain the basic mechanism of handling Buffers and Channels in Java with an example.